OUTPLAYED TURNS ARE NOT NOISE — the click-censoring fix

Version 3.41.0 · 2026-08-05 · Ordered by Will: *"i def dont like just tossing turns because they got outplayed with a move liek encore or follow me, these are the basis of vgc man."* He is right, and the literature says he is right. This document is the proposal AND the dispatch spec; MEASURE implements it, ENGINE consults on protocol detection.

---

1. The problem, stated precisely

The policy fits learn from human clicks reconstructed out of replay logs. The log records what **happened**; the fit needs what was **clicked**. Whenever an opponent's play makes those differ, today's matcher does one of two things, and both are wrong in different ways:

failure mode	mechanism examples	what the matcher does today	statistical name		
executed **target** ≠ clicked target	Follow Me, Rage Powder	**drops the turn** — counted in `unmatchedClicks` (2.92%) / `fit_joint` unmatched (5,555 + 18 ambiguous of 101,459)	biased censoring		
executed **move** ≠ clicked move, but legal-looking	Encore's application turn (Prankster Encore → victim executes last move, not its click)	**keeps it, labeled wrong** — counted nowhere	label noise		
click erased entirely	flinch, full paralysis, sleep (`\	cant\	` lines carry no move)	drops — correctly, the click is unrecoverable	genuine missingness

Two properties make this urgent rather than cosmetic:

The censoring is MNAR — Missing Not At Random (Rubin, 1976). A turn is dropped *because* the opponent's redirection worked. Rubin's taxonomy is exactly the frame: data Missing Completely At Random costs only sample size; data missing **as a function of the outcome itself** biases every estimate computed on the survivors. We are not losing 5% of turns at random — we are systematically deleting the turns where a core VGC mechanic succeeded, then fitting a policy on the sanitized remainder and playing it in the unsanitized game.

The label noise is silent and adversarially placed. The Encore application turn passes every counter we have: the executed move was on the victim's legal menu, so it "matches," and a click that never happened enters the training set — concentrated precisely on turns where the opponent outplayed the player. Learning with mislabeled examples is strictly harder than learning with missing ones (Natarajan, Dhillon, Ravikumar & Tewari, 2013): a dropped turn costs information; a poisoned turn *supplies* misinformation.

2. What the literature says to do instead

The redirection case is textbook **partial-label learning** (Cour, Sapp & Taskar, JMLR 2011, "Learning from Partial Labels"): each training instance comes with a *candidate set* of labels guaranteed to contain the true one. A redirected attack has a candidate set of exactly two — the visible final target and the target the player may have aimed at before the drag. The correct likelihood contribution is not "pick one" and not "drop it"; it is the **marginal over the candidate set**:

```
ℓ_turn(w) = log  Σ_{c ∈ C_turn}  P_w(c | board, choice set)
```

For our conditional logit (McFadden, 1974 — the estimator MAG already uses), this marginal is a log-sum-exp of affine functions minus the usual normalizer: a difference of convex functions, so the standard fitting move is **EM** (Dempster, Laird & Rubin, 1977) — E-step: distribute each partial turn's weight across its candidates in proportion to the current model's probabilities; M-step: the ordinary conditional-logit fit on weighted rows, which is the code we already have. (Equivalently CCCP, Yuille & Rangarajan 2003; EM is the implementation we choose because the M-step is literally `fit_policy`'s existing inner loop with row weights.)

Cour et al.'s identifiability condition matters here and is satisfied: partial labels are informative when the candidate sets vary — and ours do, because which Pokémon carries Follow Me varies across games, so no pair of moves is *systematically* confounded.

The Encore case is not partial-label — the true click is **gone**. The fix is detection, not inference: the protocol announces the coercion (`|-start|POKEMON|Encore` before the victim's action; forced actions generally carry `[from]` provenance, and forced switches are `|drag|` rather than `|switch|`). A detected coerced action is reclassified from "label" to "not-a-click" — removed from the labeled set and **counted**, converting silent poison into loud, honest missingness. The flinch/sleep/paralysis class stays dropped (the click is unrecoverable), but each slot drops alone: today an unprovable slot in a pair turn takes its provable partner down with it, which is pure waste.

(For completeness: state-only turns are not information-free — imitation-from-observation exists (Torabi, Warnell & Stone, 2018) — but that is a research direction, not this fix.)

3. The fix, in four stages, each with its gate

Stage A — the census instrument. A classifier over the protocol's own annotations (`[from]` , `|-activate|` , `|-start|Encore` , `|cant|` , `|drag|`) that labels every human action in the corpus: CLEAN_CLICK / PARTIAL (with its candidate set) / COERCED / ERASED. Named test case from Will: priority moves stopped by a blocking ability (Sneasler's Fake Out into a switched-in Farigiraf's Armor Tail) — whether the intended slot survives in the fail line is a fact about the protocol, to be read from real logs in the store, and it decides PARTIAL vs ERASED for that whole class. Artifact `data/click-censoring-census.json` with per-mechanism counts. **The list of mechanisms is derived from the protocol annotations, not typed from memory** — the whole lesson of this repo is that hand lists rot. *Gate: a synthetic log with a planted Encore application and a planted redirection must classify COERCED and PARTIAL respectively; shown failing first by running the classifier without the Encore rule.*

Stage B — stop the poison. The matcher consumes the census: COERCED actions leave the labeled set, counted in a new `coercedActions` counter with a declared ceiling in the degradation budgets. This is the priority stage because wrong labels are worse than missing ones (§2). *Gate: refitting after Stage B on the same corpus must change the label count by exactly the census's COERCED count; the planted-log test from Stage A wired into* `tests/` .

Stage C — keep the outplayed turns. PARTIAL actions enter the fit under the marginal likelihood via EM row-weighting; pair turns keep their provable slot regardless of the partner. *Gate: the EM harness is validated on synthetic data first — generate clicks from KNOWN weights, censor them with the real censoring process, and require the fit to recover the planted weights within tolerance where the naive (dropping) fit demonstrably does not. That is the known-bad-input demonstration, and it doubles as the measurement of how much bias Stage C removes.*

Stage D — refit and measure, paired. Both layers refit; held-out paired comparison against tonight's four-channel weights, same protocol as `data/sheet-channel-value.json` (game-clustered bootstrap, noise floor stated). Plus the number this whole fix exists for: **behavior on the outplayed-turn class** — on held-out turns where redirection/Encore is active, the model's agreement with human clicks, before vs after. If learning from outplayed turns changes how the bot plays outplayed turns, that is the point, proven.

4. What this does NOT claim

- No claim that the recovered turns improve overall top-1 — the class is ~5% of turns; the honest expectation is a real likelihood gain concentrated on redirection/Encore contexts, and Stage D's class-conditional number is the one to watch, not the corpus-wide average.
- ERASED clicks stay lost. The fix's promise is: **never keep a wrong label, never drop a provable one, count everything that remains.**
- The `turnsDropped` ceiling question (5.49%) is superseded rather than resolved: Stages B–C change what "dropped" means, so the budget gets re-derived from the new counters with its granularity stated — the re-cut Will already has on his desk folds into this.

5. References

- Rubin, D.B. (1976). Inference and missing data. *Biometrika* 63(3).
- Dempster, A.P., Laird, N.M., Rubin, D.B. (1977). Maximum likelihood from incomplete data via the EM algorithm. *JRSS B* 39(1).
- Cour, T., Sapp, B., Taskar, B. (2011). Learning from partial labels. *JMLR* 12.
- Natarajan, N., Dhillon, I., Ravikumar, P., Tewari, A. (2013). Learning with noisy labels. *NeurIPS 26*.
- Yuille, A., Rangarajan, A. (2003). The concave-convex procedure. *Neural Computation* 15(4).
- McFadden, D. (1974). Conditional logit analysis of qualitative choice behavior.
- Torabi, F., Warnell, G., Stone, P. (2018). Behavioral cloning from observation. *IJCAI*.